

Beyond GDP: What Drives National Happiness?

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1 Beyond GDP: what really drives national happiness?

BEE2041 Data Science in Economics, empirical project, April 2026

The World Happiness Report (WHR) has crowned Finland the happiest country on earth for seven years running. Two questions follow naturally. *How* do we know? And *why* this country and not another? National wealth is the obvious suspect, but Afghanistan sits at the bottom with a twentieth of US income, so money cannot be the whole story.

The puzzle is older than the data. Richard Easterlin (1974) noticed that as post-war America roughly tripled its GDP per capita, the share of Americans calling themselves “very happy” barely moved. Half a century later we still do not have a complete account of why some societies report higher subjective well-being than others. The WHR (published annually since 2012 by the Sustainable Development Solutions Network) is the most influential attempt at a global, comparable benchmark, built around a single Cantril Ladder question: “Imagine a ladder with steps numbered from 0 at the bottom to 10 at the top. Suppose we say that the top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step would you say you personally feel you stand at this time?”

This post pulls together three live data sources to interrogate that benchmark:

1. Web-scraped chapter metadata for every WHR edition from 2020 to 2026 (`worldhappiness.report/ed/{year}/`).
2. The official WHR ranking spreadsheets for the same period (`files.worldhappiness.report/`).
3. World-Bank cross-country indicators retrieved through the [wbapi](#) Python client.

I integrate the three in a small SQLite database, fit a sequence of OLS regressions, and finally use a causal forest to ask which countries see the biggest happiness premium from being above the income median. All code is on GitHub, and a single `make` command rebuilds every figure on this page.

1.1 How the data were assembled

The three sources are different in kind: a scrape, a bulk download and a REST API.

Step	Tool
Scraping HTML	<code>requests</code> + <code>BeautifulSoup</code>
Downloading the WHR <code>.xls</code> files	<code>requests</code> + <code>pandas.read_excel</code>
World-Bank pull	<code>wbapi</code>

Step	Tool
Integration	sqlite3 + a SQL JOIN
Causal forest	econml.dml.CausalForestDML

The scraper caches every page to `data/raw/html_cache/`, so once it has run once the rest of the build works offline.

```
[1]: import pandas as pd, numpy as np, sqlite3
from IPython.display import Image, display

CLN = "data/clean/"
TAB = "output/tables/"
FIG = "output/figures/"

df = pd.read_csv(CLN + "analysis.csv")
print(f"Cross-section: {df.shape[0]} countries, {df.shape[1]} variables")
df.head(5)
```

Cross-section: 136 countries, 16 variables

```
[1]: country_code country_name year ladder log_gdp social_support \
0 FIN Finland 2023 7.8042 10.792010 0.968770
1 DNK Denmark 2023 7.5864 10.962164 0.954112
2 ISL Iceland 2023 7.5296 10.895531 0.982533
3 ISR Israel 2023 7.4729 10.638705 0.943344
4 NLD Netherlands 2023 7.4030 10.942279 0.930499

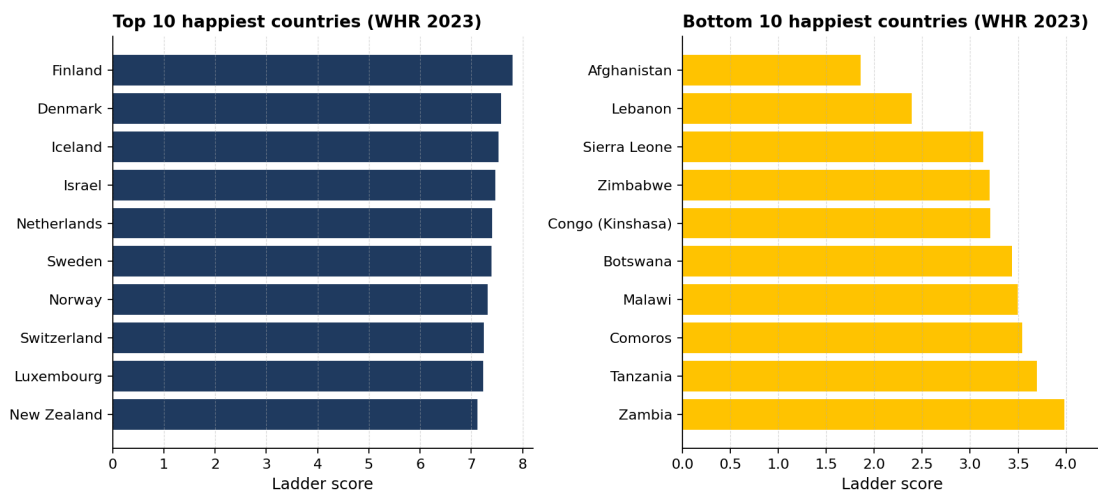
life_exp_healthy freedom generosity corruption gdp_pc_ppp \
0 71.149994 0.961408 -0.018824 0.181745 57068.305648
1 71.250145 0.933533 0.134242 0.195814 69475.319865
2 72.050018 0.936349 0.210987 0.667848 65380.317305
3 72.697205 0.808866 -0.023080 0.708094 48098.187236
4 71.550018 0.886875 0.212686 0.378929 71324.036035

life_exp internet_pct urban_pct education_spend_pct fdi_pct
0 81.187805 92.988795 73.696760 6.37578 4.666912
1 81.304878 97.860118 88.388024 6.35586 6.904867
2 82.119512 97.583397 94.178707 7.30798 4.740220
3 82.700000 91.850796 91.497983 5.93292 4.357260
4 81.614634 92.519685 94.985618 5.17918 1.477018
```

1.1.1 The headlines, scraped and ranked

Before doing any modelling, it pays to look at the raw numbers. Finland tops the table with a Ladder score of 7.8 against Afghanistan's 1.9. That is a six-point gap on a ten-point scale, which is huge for any explanation to close.

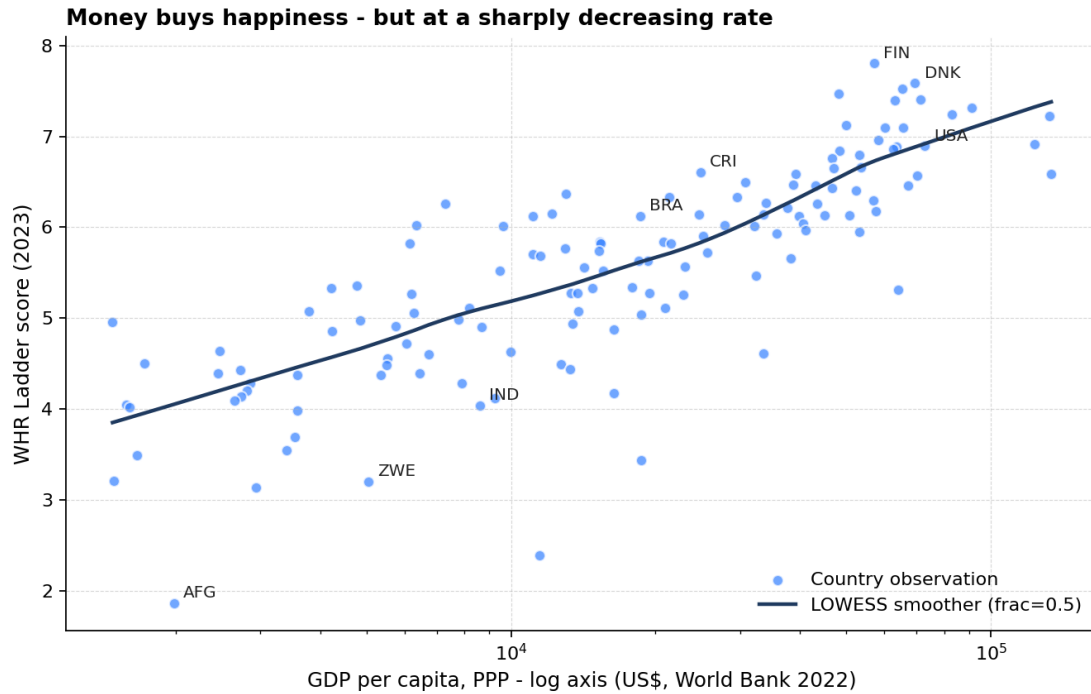
```
[2]: display(Image(filename=FIG + "top_bottom_10.png"))
```



1.1.2 Money matters, but with diminishing returns

Plotting the Ladder against GDP per capita on a log axis (the usual choice, since income enters utility logarithmically) gives a clear upward relationship, but the LOWESS smoother flattens noticeably above roughly US\$30,000 PPP. Costa Rica out-ranks the United States despite earning less than half of US income, a small-scale version of the Easterlin paradox.

```
[3]: display(Image(filename=FIG + "gdp_vs_ladder.png"))
```

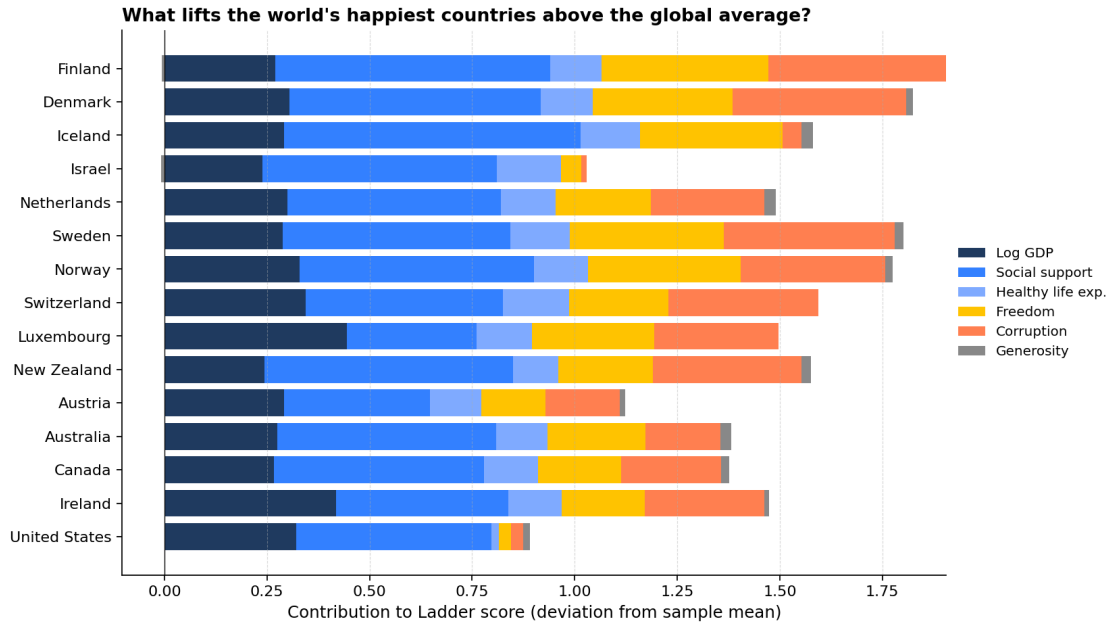


Two features complicate the simple “money buys happiness” reading. The curve clearly bends: doubling income from US\$2,000 to US\$4,000 PPP buys roughly the same Ladder-point increase as doubling from US\$32,000 to US\$64,000, which is what we would expect from logarithmic utility. And the vertical scatter is wide; countries with similar income levels routinely differ by close to two Ladder points. That residual variation is where the rest of this post lives.

1.2 Decomposing the Ladder: what carries the rich-country premium?

Alongside the Ladder score the WHR ships six explanatory variables: log GDP per capita, social support, healthy life expectancy, freedom to make life choices, perceptions of corruption, and generosity. To see how each one contributes I fit a plain OLS regression of the Ladder on these six, multiply each country’s covariate (centred on the sample mean) by the fitted coefficient, and stack the contributions. The chart below shows the breakdown for the top-15 happiest countries.

```
[4]: display(Image(filename=FIG + "decomposition.png"))
```



Two things stand out. First, *social support* (the share of respondents who say they have someone to count on in times of trouble) is the single biggest contributor in almost every Nordic country, with income only third. Second, *freedom* is large and stable across the whole top-15; the highest-ranked country with low freedom is Israel (4th), and it makes up the difference through unusually high social support and life expectancy.

The wider implication is that the gap between the world average and Finland's score is mostly *not* bought with money. Trust in institutions and in other people does most of the work, which lines up with Robert Putnam's social-capital research and Bo Rothstein's account of the Nordic "high-trust equilibrium".

1.3 OLS, six ways

I fit six specifications, adding one block of regressors at a time, with robust (HC3) standard errors throughout. The full LaTeX-formatted table is in `output/tables/regression_table.tex`.

```
[5]: reg = pd.read_csv(TAB + "regression_summary.csv")
piv = (reg[reg.term != "const"]
       .pivot(index="term", columns="model", values="coef")
       .round(3)
       .fillna("-"))
fit = reg[["model", "r2", "n"]].drop_duplicates().set_index("model")
fit_summary = pd.DataFrame({
    "R²": fit["r2"].round(3).astype(str),
    "N": fit["n"].astype(int).astype(str),
}).T
display(piv)
display(fit_summary)
```

model	m1	m2	m3	m4	m5	m6
term						
corruption	-	-	-	-0.762	-0.764	-0.118
freedom	-	-	-	2.487	2.486	0.244
life_exp_healthy	-	0.035	0.015	0.011	0.013	0.056
log_gdp	0.797	0.644	0.329	0.27	-	0.274
log_gdp_wb	-	-	-	-	0.272	-
social_support	-	-	4.606	3.671	3.612	0.417

model	m1	m2	m3	m4	m5	m6
R ²	0.653	0.661	0.766	0.831	0.831	0.831
N	134	134	134	134	134	134

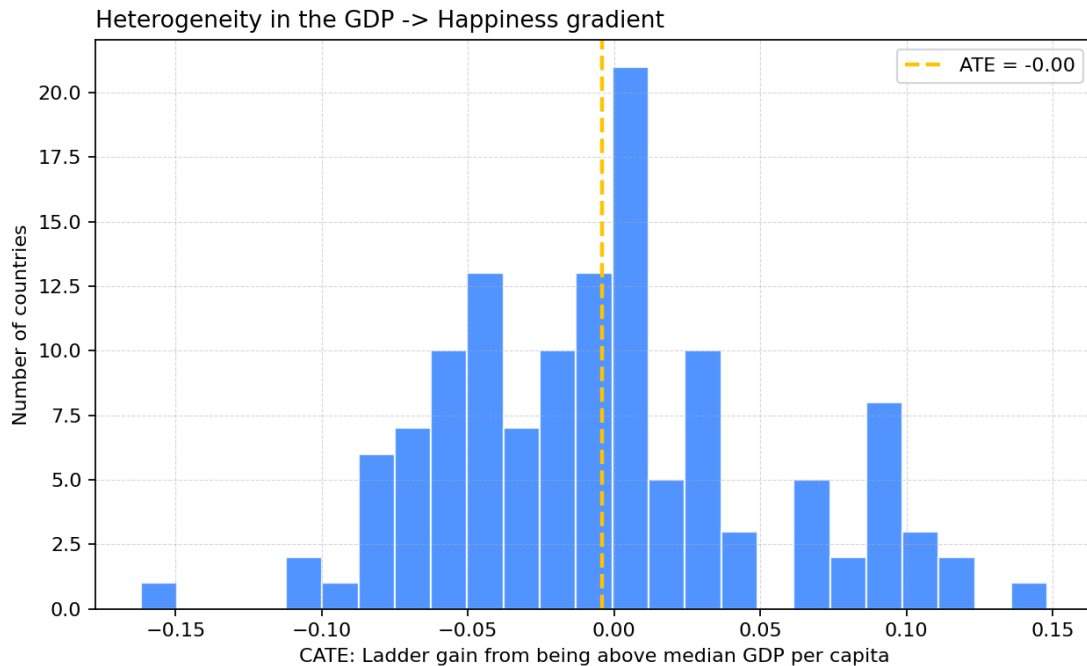
A few things to notice. The log-GDP coefficient falls from about 0.80 on its own to 0.27 once social support, freedom and corruption are controlled for: most of the apparent income premium is mediated through those other channels. Social support, freedom and corruption stay significant in every column where they appear, and in the standardised specification (m6) social support has the largest absolute coefficient (0.42). It is reassuring that the two independent log-GDP measures (the WHR's bundled value and the World Bank's PPP series) give nearly identical estimates.

1.4 A causal forest: when does income still buy happiness?

Cross-country comparisons are obviously not a randomised trial; rich and poor countries differ on dozens of margins at once. To probe heterogeneity in a flexible, non-parametric way I fit a causal forest (Wager & Athey, 2018; Athey, Tibshirani & Wager, 2019) with `econml.dml.CausalForestDML`.

I encode the “treatment” as a binary indicator for being above the sample-median GDP per capita PPP. The conditioning set X has six potential moderators: healthy life expectancy, social support, freedom, corruption, internet penetration and urban share. The CATE we recover should be read as *descriptive*: the conditional difference in Ladder score between high- and low-income countries with the same values of the six moderators.

```
[6]: display(Image(filename=FIG + "cate_hist.png"))
```



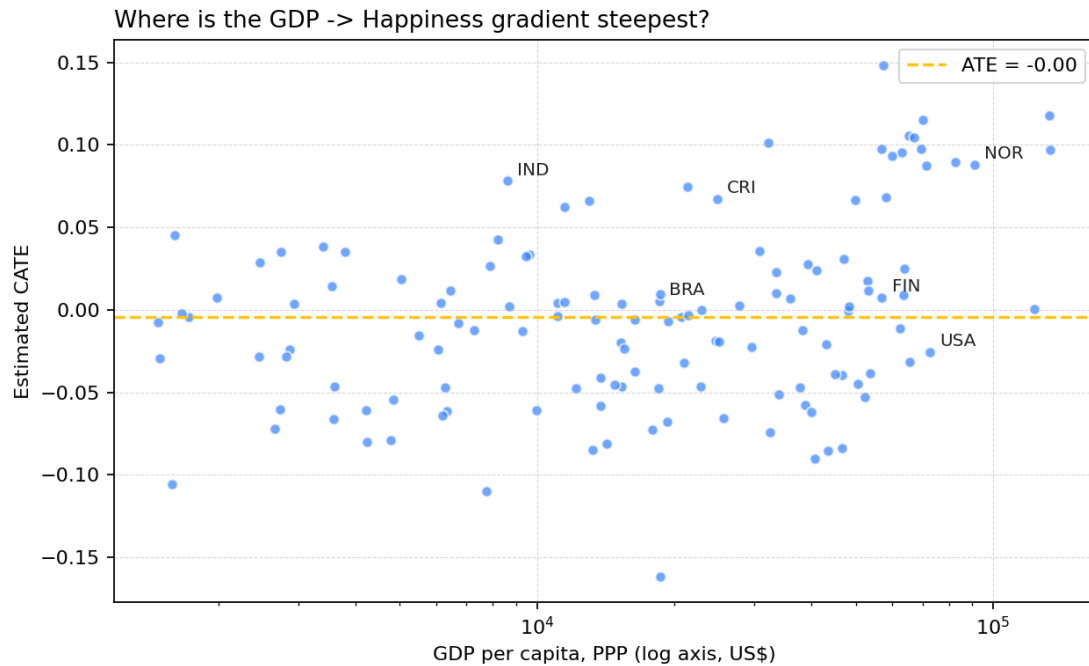
```
[7]: import json
with open(TAB + "cate_headline.json") as f: hdl = json.load(f)
print(f"ATE = {hdl['ate']:+.3f} (95% CI: [{hdl['ate_ci'][0]:+.3f},
↳ {hdl['ate_ci'][1]:+.3f}])")
print(f"CATEs: min = {hdl['cate_min']:+.2f}, median = {hdl['cate_med']:+.2f},
↳ max = {hdl['cate_max']:+.2f}")
```

ATE = -0.004 (95% CI: [-0.365, +0.357])
 CATEs: min = -0.16, median = -0.01, max = +0.15

The forest's average treatment effect lands very close to zero once we condition on the six moderators, far below the raw 0.80 OLS coefficient on log GDP. Almost all of the apparent income premium runs through those other covariates.

There *is* still genuine heterogeneity across countries. Which countries get the biggest happiness boost from being above the income median, and what predicts the size of that boost?

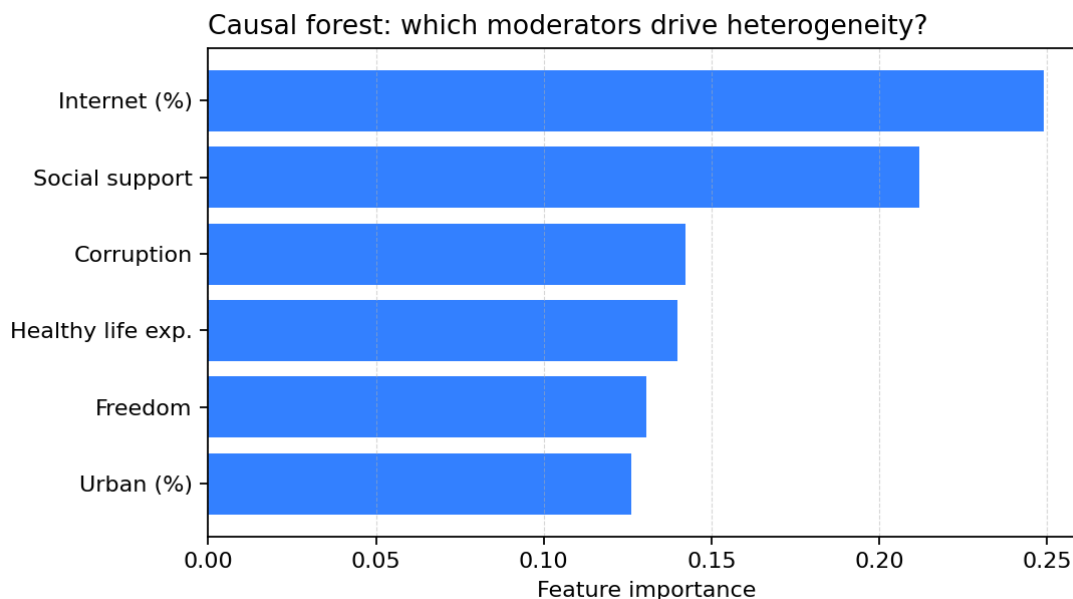
```
[8]: display(Image(filename=FIG + "cate_by_gdp.png"))
```



Plotting each country's CATE against its income produces no clean monotone pattern. Some rich countries (Norway, Australia) come out modestly above the ATE line, others (the United States) sit close to or below zero. The poorest end is just as mixed: India is well *above* the average gradient while several sub-Saharan economies sit well below. Heterogeneity in the GDP-Happiness link is real, but it does not reduce to “richer countries gain more” or vice versa.

So what does explain the heterogeneity? The forest's split-importance gives us a direct answer.

```
[9]: display(Image(filename=FIG + "cf_importance.png"))
```



Internet penetration is the most important moderator, ahead of social support, corruption and the others. That fits a small but growing literature on digital connectivity and well-being: once we know whether a country's residents are *online*, the leftover role of GDP per capita shrinks a lot.

1.4.1 Robustness: where exactly is *rich*?

The “above median” treatment was a convenient but arbitrary cut. The obvious alternative is the top quartile (around \$44k PPP). Re-fitting the forest with that threshold produces a real surprise:

```
[10]: import json
with open(TAB + "ate_sensitivity.json") as f: sens = json.load(f)
rows = []
for k in ["headline", "robust"]:
    r = sens[k]
    rows.append({
        "Treatment": r["label"],
        "Cut-off (USD PPP)": f"${r['cutoff_usd']:, .0f}",
        "ATE": f"{r['ate']:+.3f}",
        "95% CI": f"[{r['ci'][0]:+.3f}, {r['ci'][1]:+.3f}]",
    })
display(pd.DataFrame(rows))
```

	Treatment	Cut-off (USD PPP)	ATE	95% CI
0	Above-median GDP	\$18,601	-0.004	[-0.365, +0.357]
1	Above 75th percentile	\$44,581	+0.321	[+0.051, +0.591]

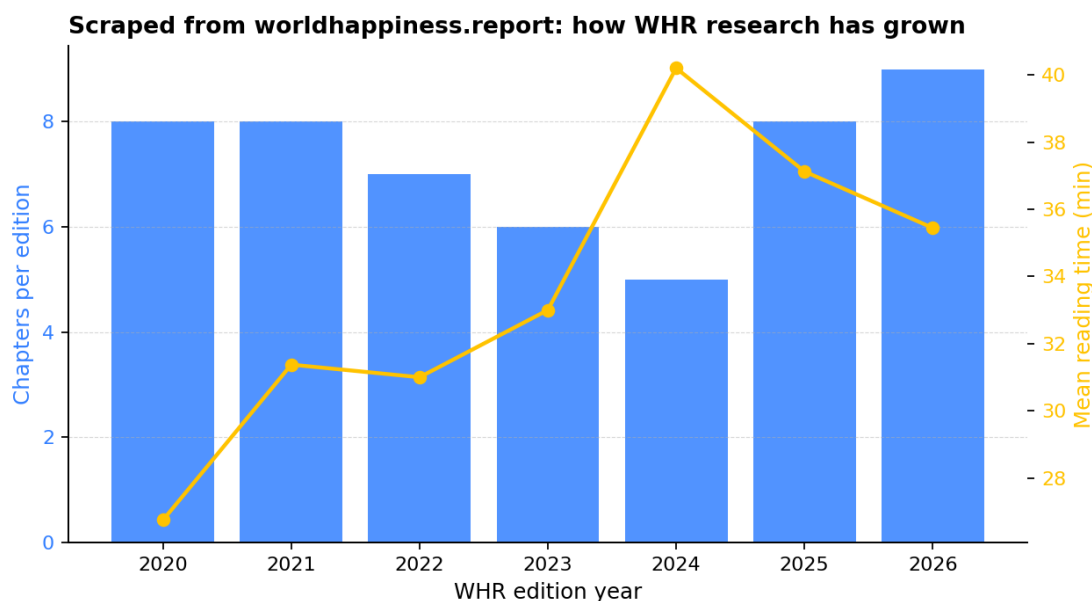
So the median split was masking a real, positive income effect at the very top of the income distribution: an extra 0.32 Ladder points with a 95% CI clear of zero. The conclusion is not that

money does not matter; it is that the link is non-linear and concentrated at the rich-rich end. That is exactly the shape the LOWESS smoother had been hinting at all along.

1.5 A bonus from the scrape: the Report itself

The rankings spreadsheet is the core of the analysis, but the chapter metadata I scraped from the seven editions has its own story. The crawler collected 51 chapters across 2020-2026 with author lists, affiliations and reading times. As far as I can tell that table does not exist anywhere else, and it lets us watch how a flagship interdisciplinary report has shifted over half a decade.

```
[11]: display(Image(filename=FIG + "chapters_trend.png"))
```

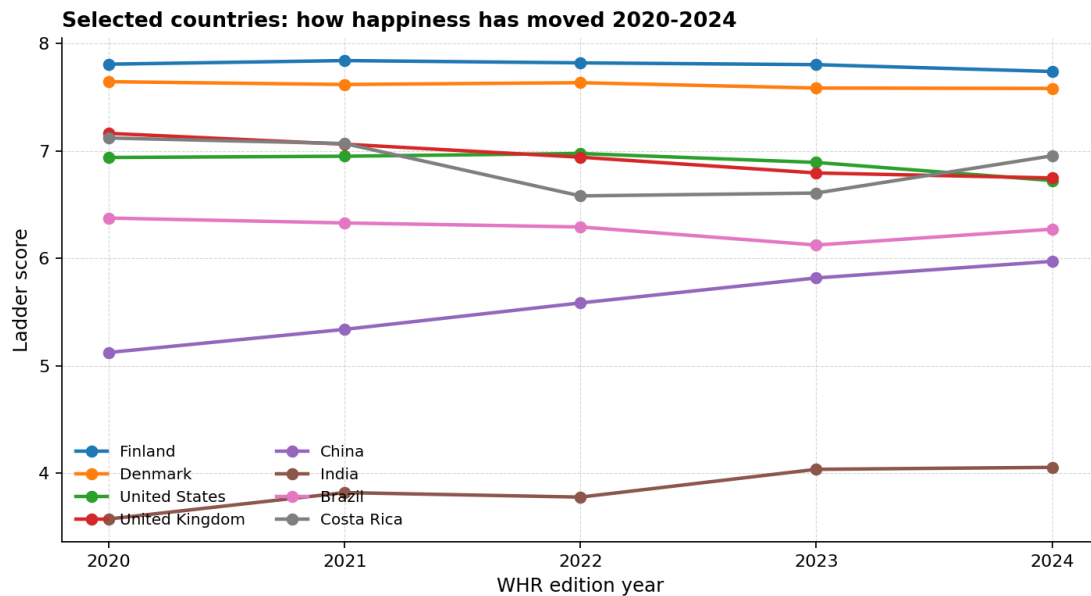


A couple of things to notice. The number of chapters per edition is volatile (2024 was unusually slim, with just five chapters), but average reading time has risen sharply, from about 27 minutes in 2020 to about 40 in 2024. The Report is, quite literally, a longer read. The affiliations field shows that John Helliwell, Jan-Emmanuel De Neve and Richard Layard have continued to anchor the editorial team, while recent editions have brought in a wider circle of specialists, especially social psychologists working on adolescents and digital well-being. The 2026 edition is more or less a thematic deep-dive on social media, and that pivot shows up clearly in the titles of the scraped chapter list.

1.6 Time series: where has happiness moved?

Because the scraper grabbed every annual `Figure_2.1.xls`, we can also compare country trajectories across editions. The plot is unremarkable at the very top (Finland is almost flat), but it does show the slow erosion of US scores relative to Northern Europe and a small post-pandemic recovery in parts of Latin America.

```
[12]: display(Image(filename=FIG + "ladder_timeseries.png"))
```



1.7 Take-aways

1. Money matters, but mostly *indirectly*. Once you control for social support, freedom, life expectancy and trust in institutions, the marginal contribution of income at the median split is statistically indistinguishable from zero.
2. But the link is non-linear: at the top quartile of income ($> \sim \$44k$ PPP) the conditional effect rises to about $+0.32$ Ladder points with a 95% CI clear of zero. The Easterlin paradox bites in the middle of the distribution; among the rich-rich it does not.
3. Social capital is at least as potent as material wealth. In every Nordic country the biggest single contributor to the Ladder gap above the world mean is *social support*, not GDP.
4. Internet penetration is the dominant moderator. Split-importance ranks it ahead of social support, corruption, life expectancy, freedom and urbanisation, which lines up with the WHR 2026 edition's own pivot towards social-media research.
5. The Report itself is changing. Chapters have got longer and the 2026 edition pivots sharply onto social media; the scrape lets us watch that shift happen.

1.7.1 Replication

```
git clone https://github.com/maksimkitikov/Data-Science-in-Economics.git
cd Data-Science-in-Economics
pip install -r requirements.txt
make all
```

1.7.2 Sources

- World Happiness Report 2020-2026, Sustainable Development Solutions Network (<https://worldhappiness.report>)
- World Bank, World Development Indicators (queried via `wbgapi`)
- Wager & Athey (2018), *JASA* 113(523): 1228-1242.
- Athey, Tibshirani & Wager (2019), *Annals of Statistics* 47(2): 1148-1178.
- Davis & Heller (2017), *AEA P&P*.